Mat2002 Tutorial 8

Junwei Zha

March 28, 2026

System of First Order Linear ODE

General system of first order ODE is

$$\begin{cases} x_1' = F_1(t, x_1, \cdots, x_n) \\ x_2' = F_2(t, x_1, \cdots, x_n) \\ \vdots \\ x_n' = F_n(t, x_1, \cdots, x_n) \end{cases}$$

where $x_1, \cdots, x_n$ are unknown functions of t
Initial condition:

$$x_1(t_0) = x_1^0, x_2(t_0) = x_2^0, \cdots, x_n(t_0) = x_n^0$$

Existence and Uniqueness Theorem

Consider the IVP. Suppose the functions $F_i, 1 \leqslant i \leqslant n$, and the partial derivatives $\frac{\partial F_i}{\partial x_j}, 1 \leq i \leq n, 1 \leq j \leq n$ are all continuous in

$$\Omega : \quad \alpha < t < \beta, \alpha_1 < x_1 < \beta_1, \cdots, \alpha_n < x_n < \beta_n$$

containing the point $(t_0, x_1^0, \cdots, x_n^0)$ Then there is an interval $I \subseteq (\alpha, \beta)$ containing t_0, such that there exists unique solution $x_1(t), \cdots, x_n(t)$ in I to IVP.

Linear System of First Order ODE

Linear system of first order ODE is

$$\begin{cases} x_1' = p_{11}(t)x_1 + \cdots + p_{11}(t)x_n + g_1(t) \\ x_2' = p_{21}(t)x_1 + \cdots + p_{2n}(t)x_n + g_2(t) \\ \qquad\qquad\qquad\qquad \vdots \\ x_n' = p_{11}(t)x_1 + \cdots + p_{nn}(t)x_n + g_n(t) \end{cases}$$

where $P_{ij}(t), g_i(t), 1 \leqslant i, j \leqslant n$ are given functions of t. The system is called homogeneous if $g_i(t) \equiv 0, 1 \leq i \leq n$. Otherwise, nonhomogeneous

Remark

We'll only focus on linear system in the rest of this chapter.

It's more convenient to rewrite the system into matrix form.

Existence and Uniqueness theorem

Theorem

Consider the IVP of linear system of first Order ODE.

$$\begin{cases} x_1' = p_{11}(t)x_1 + \cdots + p_{11}(t)x_n + g_1(t) \\ x_2' = p_{21}(t)x_1 + \cdots + p_{2n}(t)x_n + g_2(t) \\ \qquad\qquad\qquad\qquad \vdots \\ x_n' = p_{11}(t)x_1 + \cdots + p_{nn}(t)x_n + g_n(t) \end{cases}$$

Suppose the functions $p_{ij}(t), g_i(t), 1 \leq i, j \leq n$ are all continuous in an open interval I containing the point to. Then there exists unique solution $x_1(t), \cdots, x_n(t)$ in I to IVP.

General Theory for Systems of First Order Linear Equations

Consider

$$\begin{cases} x_1' = p_{11}(t)x_1 + \cdots + p_{1n}(t)x_n + g_1(t) \\ x_2' = p_{21}(t)x_1 + \cdots + p_{2n}(t)x_n + g_2(t) \\ \qquad\qquad\qquad\qquad \vdots \\ x_n' = p_{n1}(t)x_1 + \cdots + p_{nn}(t)x_n + g_n(t) \end{cases}$$

Denote $\vec{x}(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix}, P(t) = \begin{pmatrix} p_{11}(t) & \cdots & p_{1n}(t) \\ \vdots & \cdots & \vdots \\ p_{n1}(t) & \cdots & p_{nn}(t) \end{pmatrix}, \vec{g}(t) = \begin{pmatrix} g_1(t) \\ \vdots \\ g_n(t) \end{pmatrix}$,

then the linear system can be written as:

$$\vec{x}' = P(t)\vec{x} + \vec{g}(t)$$

Homogeneous equation: $\vec{x}' = P(t)\vec{x}$

General Theory for Systems of First Order Linear Equations

A very important remark:
nth order linear equation can be regarded as a special case of first order linear system:

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_0(t)y = g(t).$$

Set $\vec{x} = \begin{pmatrix} y \\ y' \\ \vdots \\ y^{(n-1)} \end{pmatrix}$, then it's equivalent to:

$$\vec{x}' = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ & \cdots & \cdots & & \\ -p_0 & -p_1 & -p_2 & \cdots & p_{n-1} \end{pmatrix} \vec{x} + \begin{pmatrix} 0 \\ 0 \\ \vdots \\ g \end{pmatrix} = p\vec{x} + \vec{g}$$

Therefore, the theory for nth order linear equation can be derived as a corollary of the theory for linear system.

Sturcture of solutions

Theorem

If $\vec{x}^{(1)}$ and $\vec{x}^{(2)}$ are two solutions of the linear homogeneous system, then the linear combination $c_1\vec{x}^{(1)} + c_2\vec{x}^{(2)}$ is also a solution for any constants c_1, c_2.

Definition

Let $\vec{x}^{(1)}, \cdots, \vec{x}^{(n)}$ be n solutions of the linear homogeneous system.Define Wronskian of $\vec{x}^{(1)}, \cdots, \vec{x}^{(n)}$ by :

$$W\left(\vec{x}^{(1)}, \cdots, \vec{x}^{(n)}\right)(t) = \det(X(t))$$

$$= \begin{vmatrix} x_1^{(1)} & \cdots & x_1^{(n)} \\ \vdots & \cdots & \vdots \\ x_n^{(1)} & \cdots & x_n^{(n)} \end{vmatrix}$$

Sturcture of solutions

Theorem

Let $\vec{x}^{(1)}, \cdots, \vec{x}^{(n)}$ be n solutions to the homogeneous system

$$\vec{x}' = P(t)\vec{x}, \quad t \in I.$$

Then the Wronskian $W\left(\vec{x}^{(1)}, \cdots, \bar{x}^{(n)}\right)(t) = ce^{\int \operatorname{tr}(P)dt}$ for some constant C. Here $\operatorname{tr}(P) = \sum_{i=1}^{n} P_{ii}$ is the trace of $P(t)$.

As a consequence, this Wronskian is either always zero, or never zero.

Remark: Just like n-th order equation case, if $\vec{x}^{(1)}, \cdots, \vec{x}^{(n)}$ are n solutions of the linear homogeneous system $\vec{x}' = P(t)\vec{x}$, then $W\left(\vec{x}^{(1)}, \cdots, \vec{x}^{(n)}\right)(t_0) \neq 0$ for some $t_0 \in I \Leftrightarrow \vec{x}^{(1)}, \cdots, \vec{x}^{(n)}$ are linearly independent in I.

Sturcture of solutions

Theorem

Let $\vec{x}^{(1)}, \cdots, \vec{x}^{(n)}$ be n solutions to the homogeneous system

$$\vec{x}' = P(t)\vec{x}, \quad t \in I.$$

Suppose $W\left(\vec{x}^{(1)}, \cdots, \vec{x}^{(n)}\right)(t_0) \neq 0$ for some $t_0 \in I$, then the general solution is $\vec{x} = c_1\vec{x}^{(1)} + \cdots + c_n\vec{x}^{(n)}$.

Furthermore, the constants $c_1, \cdots, c_n$ can be uniquely determined by the initial condition. In this case, we say $\vec{x}^{(1)}, \cdots, \vec{x}^{(n)}$ form a fundamental set of solutions.

Exercise

Verify that

$$y = c_1 \begin{bmatrix} e^x \cos x \\ e^x \sin x \end{bmatrix} + c_2 \begin{bmatrix} -\sin x \\ \cos x \end{bmatrix}$$

is the general solution of the following ODE system:

$$y' = \begin{bmatrix} \cos^2(x) & \frac{1}{2}\sin 2x - 1 \\ \frac{1}{2}\sin 2x + 1 & \sin^2(x) \end{bmatrix} y.$$

Exercise

Solve

$$y' = \begin{bmatrix} 1 & 1 \\ 0 & \frac{1}{x} \end{bmatrix} y.$$

Hint: Don't forget the actual mean of matrix form.

Solution: $y = c_1 \begin{bmatrix} e^x \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -x-1 \\ x \end{bmatrix}$

Homogeneous Linear System with constant Coefficients

Consider $\vec{x}' = A\vec{x}$ where A is a $n \times n$ real-valued constant matrix.
Recall: $n = 1$. $x' = rx, x = e^{rt}$ is a solution.
Expect: general n, $\vec{x}' = A\vec{x}$, $\vec{x} = e^{rt}\vec{\xi}$, $\vec{\xi} \neq \overrightarrow{0}$: constant vector.
Substitute $\vec{x} = e^{rt}\vec{\xi}$ into the equation,

$$e^{rt} r\vec{\xi} = e^{rt} A\vec{\xi}$$
$$\Leftrightarrow \quad A\vec{\xi} = r\vec{\xi}, \vec{\xi} \neq 0.$$

It means r is an eigenvalue, $\vec{\xi}$ is the corresponding eigenvector.

Procedure to find fundamental set of solution

Procedure to find fundamental set of solution: $\vec{x}' = A\vec{x}$.
(i) Find all eigenvalues of A : $\lambda_1, \cdots, \lambda_n$ (may be equal)
(ii) Find the corresponding n linearly independent eigenvectors $\vec{\xi}_1, \cdots, \vec{\xi}_n$ (if they exist) $\Rightarrow \left\{ e^{\lambda_1 t}\vec{\xi}_1, e^{\lambda_2 t}\vec{\xi}_2, \cdots, e^{\lambda_n t}\vec{\xi}_n \right\}$ form a fundamental set of solutions.
(iii) If there are not n linearly independent eigenvectors, more solutions are needed.

Real eigenvalue

Case I. No repeated eigenvalue i.e. $\lambda_1 \neq \lambda_2 \neq \cdots \neq \lambda_n$, algebraic multiplicity = geometric multiplicity = 1. then there must exist n linearly independent eigenvectors $\vec{\xi}_1, \cdots, \vec{\xi}_n$

Example

Find the general solution of $\vec{x}' = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \vec{x}$.

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 1 \\ 4 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 - 4 = 0, \quad \lambda_1 = 3, \quad \lambda_2 = -1.$$

For $\lambda_1 = 3$, $(A - 3I)\vec{\xi}_1 = 0$,

$$A - 3I = \begin{pmatrix} -2 & 1 \\ 4 & -2 \end{pmatrix} \to \begin{pmatrix} -2 & 1 \\ 0 & 0 \end{pmatrix} \vec{\xi}_1 = \binom{1}{2}.$$

For $\lambda_2 = -1$, $(A + I)\xi_2 = 0$.

$$A + I = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} \to \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix}, \vec{\xi}_2 = \binom{-1}{2}$$

So general solution is $\vec{x} = c_1 e^{3t}\binom{1}{2} + c_2 e^{-t}\binom{-1}{2} = \binom{c_1 e^{3t} - c_2 e^{-t}}{2c_1 e^{3t} + 2c_2 e^{-t}}, \quad \forall c_1, c_2$

Example II

Solve

$$\vec{x}' = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix} \vec{x}$$

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 1 \\ 0 & 0 & 3-\lambda \end{vmatrix} = (1-\lambda)(2-\lambda)(3-\lambda) = 0 \Rightarrow$$

$\lambda = 1, 2, 3.$

For $\lambda_1 = 1$, $\vec{\xi}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ For $\lambda_2 = 2$, $\vec{\xi}_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$

For $\lambda_3 = 3, \vec{\xi}_3 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$

$$\Rightarrow \vec{x} = c_1 e^t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + c_3 e^{3t} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

Repeated eigenvalue. alg. mul.= geo. mul.

Example

$$\vec{x}' = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \vec{x}$$

Sol: $\det(A - \lambda I) = (2 - \lambda)(\lambda + 1)^2$. $\lambda_1 = 2$, unrepeated, $\vec{\xi}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

$\lambda_2 = -1$, repeated, $\vec{\xi}_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$, $\vec{\xi}_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.

General solution is $\vec{x} = c_1 e^{2t} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c_3 e^{-t} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.

Rmk: This is actually the same as the previous one, i.e. you can find enough eigenvector.

Repeated eigenvalue, without n linearly independent eigenvectors

There exist some eigenvalue, with geometric multiplicity $<$ algebraic multiplicity. We need to find more solutions to form a fundamental set.
For simplicity, we first consider eigenvalue λ with

$$\begin{cases} \text{algebraic multiplicity } 2 \\ \text{geometric multistiaity } 1 \end{cases}$$

Denote $\vec{\xi}$ to be the eigenvector, we have one solution $\vec{\chi}^{(1)} = e^{\lambda t}\vec{\xi}$.
To find another solution, substitute $\vec{x}^{(2)} = e^{\lambda t}\vec{\eta} + te^{\lambda t}\vec{\xi}$ gives

$$\lambda e^{\lambda t}\vec{\eta} + \lambda t e^{\lambda t}\vec{\xi} + e^{\lambda t}\vec{\xi} = e^{\lambda t}A\vec{\eta} + \lambda t e^{\lambda t}\vec{\xi}$$
$$\vec{\xi} = (A - \lambda I)\vec{\eta}$$

Hence two solutions are $\vec{x}^{(1)} = e^{\lambda t}\vec{\xi}, \vec{x}^{(2)} = e^{\lambda t}(\vec{\eta} + t\vec{\xi})$, where $\vec{\xi}$ is the eigenvector, $\vec{\eta} \neq 0, s.t.(A - \lambda I)\vec{\eta} = \vec{\xi}$.
Remark: It's easy to check $(A - \lambda I)^2\eta = 0$, and η is called generalized eigenvector.

Example

$$\vec{x}' = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix} \vec{x}$$

Sol: $|A - \lambda I| = \begin{vmatrix} 1-\lambda & -1 \\ 1 & 3-\lambda \end{vmatrix} = (\lambda - 2)^2 = 0$, $\lambda = 2$, with algebraic multipliaty 2.

For $\lambda = 2$, $\quad A - 2I = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$

Only one eigenvector $\vec{\xi} = \binom{1}{-1}, \vec{x}(1) = e^{2t}\binom{1}{-1}$, geometric multiplicity 1 .

Let $(A - \lambda I)\vec{\eta} = \vec{\xi}$, $\quad \vec{\eta}$ is the generalized eigenvector.

$$\left(\begin{array}{cc|c} -1 & -1 & 1 \\ 1 & 1 & -1 \end{array}\right) \to \left(\begin{array}{cc|c} -1 & -1 & 1 \\ 0 & 0 & 0 \end{array}\right),$$

$\vec{\eta} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} + c\begin{pmatrix} 1 \\ -1 \end{pmatrix}$: $c\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ can be absorbed in $\vec{x}^{(1)} = e^{\lambda t}\vec{\xi}$.

$$\vec{x}_2 = e^{\lambda t}(\vec{\eta} + t\vec{\xi}) = e^{2t}\begin{pmatrix} 0 \\ -1 \end{pmatrix} + te^{2t}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

General sol: $\vec{x} = c_1\vec{x}^{(1)} + c_2\vec{x}^{(2)} = c_1 e^{2t}\binom{1}{-1} + c_2 e^{2t}\binom{0}{-1} + c_2 te^{2t}\binom{1}{-1}$.

Example II

$$\vec{x}' = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 1 \\ 2 & 0 & 1 \end{pmatrix} \vec{x}$$

Sol: $\det(A - \lambda I) = (\lambda - 3)(\lambda + 1)^2$. $\lambda_1 = 3, \quad \lambda_2 = \lambda_3 = -1.$

For $\lambda_1 = 3, \quad \vec{\xi}_1 = \begin{pmatrix} 1 \\ \frac{1}{2} \\ 1 \end{pmatrix}, \quad \vec{x}^{(1)} = e^{\lambda_1 t}\vec{\xi}_1 = e^{3t} \begin{pmatrix} 1 \\ \frac{1}{2} \\ 1 \end{pmatrix}.$

For $\lambda_2 = -1$, algebraic multiplicity 2 . geometric multiplicity 1,

$\vec{\xi}_2 = \begin{pmatrix} -1 \\ \frac{1}{2} \\ 1 \end{pmatrix}, \vec{x}^{(2)} = e^{-t} \begin{pmatrix} -1 \\ \frac{1}{2} \\ 1 \end{pmatrix}$

To find one more solution associated with $\lambda_2 = -1$, solve $(A - \lambda_2 I)\,\eta = \vec{\xi}_2$,

$$\left(\begin{array}{ccc|c} 2 & 2 & 1 & \frac{1}{2} \\ 1 & 0 & 1 & -1 \\ 2 & 0 & 2 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & 1 & \frac{1}{2} \\ 0 & 1 & -\frac{1}{2} & -1 \\ 0 & 0 & 0 & 0 \end{array}\right), \quad \vec{\eta} = \begin{pmatrix} \frac{1}{2} \\ -1 \\ 0 \end{pmatrix}$$

Example II

so $\vec{x}^{(3)} = e^{\lambda_2 t}\left(\vec{\eta} + t\vec{\xi}_2\right) = e^{-t}\begin{pmatrix} \frac{1}{2} \\ -1 \\ 0 \end{pmatrix} + te^{-t}\begin{pmatrix} -1 \\ \frac{1}{2} \\ 1 \end{pmatrix}.$

General solution: $\vec{x} = c_1\vec{x}^{(1)} + c_2\vec{x}^{(2)} + c_3\vec{x}^{(3)}$.

More complicated case

We will take 3×3 systems as example to explain how to use generalized eigenvector to find more solutions if geometric multiplicity $<$ algebroare multiplicity. There are 3 situations:
Case 1: Geometric multiplicity $= 1$, Algebraic multiplicity $= 2$
Case 2: Geometric multiplicity $= 1$, Algebraic multiplicity $= 3$
Case 3. Geometric multiplicity $= 2$, Algebraic multiplicity $= 3$

Case 1 has been treated in previous example.
Case 2. λ, eigenvector $\vec{\xi} \Rightarrow$ one solution $\vec{x}^{(1)} = e^{\lambda t}\vec{\xi}$. we need to find two more solutions. Firstly, solve $(A - \lambda I)\vec{\eta} = \vec{\xi}$ to find one generalized eigenvector $\vec{\eta}$, then $\vec{x}^{(2)} = e^{\lambda t}(\vec{\eta} + t\vec{\xi})$. Secondly. let's use $\vec{x}^{(3)} = e^{\lambda t}\left(\vec{\theta} + t\vec{\eta} + \frac{t^2}{2}\vec{\xi}\right)$

$$\vec{x}^{(3)} = A\vec{x}^{(3)} \Leftrightarrow \quad (A - \lambda I)\vec{\theta} = \vec{\eta}$$

Hence, to find the 3 solutions, we need:

$$(A - \lambda I)\vec{\xi} = 0, \quad (A - \lambda I)\vec{\eta} = \vec{\xi}, \quad (A - \lambda I)\vec{\theta} = \vec{\eta}$$

More complicated case

Rmk:

$(A-\lambda I)^2\vec{\eta}=0$. $\vec{\eta}$ is called generalized eigenvector of rank 2 .

$(A-\lambda I)^3\vec{\theta}=0$, $\vec{\theta}$ is called generalized eigenvector of rank 3 .

$$\begin{cases} \vec{x}^{(1)} = e^{\lambda t}\vec{\xi} \\ \vec{x}^{(2)} = e^{\lambda t}(\vec{\eta}+t\vec{\xi}) \\ \vec{x}^{(3)} = e^{\lambda t}\left(\vec{\theta}+t\vec{\eta}+\frac{t^2}{2}\vec{\xi}\right) \end{cases}$$

Example

$$\vec{x}' = \begin{pmatrix} -7 & -5 & -3 \\ 2 & -2 & -3 \\ 0 & 1 & 0 \end{pmatrix}$$

Sol: $\det(A - \lambda I) = -(\lambda + 3)^3 = 0 \Rightarrow \lambda = -3, -3, -3$. AM is 3

$$A + 3I = \begin{pmatrix} -4 & -5 & -3 \\ 2 & 1 & -3 \\ 0 & 1 & 3 \end{pmatrix} \to \begin{pmatrix} 2 & 1 & -3 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \text{one eigenvector}$$

$$\vec{\xi} = \begin{pmatrix} 3 \\ -3 \\ 1 \end{pmatrix}$$

Compute $(A + 3I)\vec{\eta} = \vec{\xi}$

$$\left(\begin{array}{ccc|c} -4 & -5 & -3 & 3 \\ 2 & 1 & -3 & -3 \\ 0 & 1 & 3 & 1 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & -3 & -2 \\ 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right) . \text{ choose } \vec{\eta} = \begin{pmatrix} 0 \\ -1 \\ \frac{2}{3} \end{pmatrix}$$

Example

Compute $(A+3I)\vec{\theta}=\vec{\eta}$

$$\left(\begin{array}{ccc|c} -4 & -5 & -3 & 0 \\ 2 & 1 & -3 & -1 \\ 0 & 1 & 3 & \frac{2}{3} \end{array}\right) \to \left(\begin{array}{ccc|c} 2 & 0 & -6 & -\frac{5}{3} \\ 2 & 2 & 0 & -\frac{1}{3} \\ 0 & 0 & 0 & 0 \end{array}\right) . \text{ choose } \vec{\theta} = \begin{pmatrix} -\frac{1}{6} \\ 0 \\ \frac{2}{9} \end{pmatrix}$$

Hence the general solution is

$$\vec{x}(t) = c_1 e^{-3t}\vec{\xi} + c_2 e^{-3t}(\vec{\eta} + t\vec{\xi}) + c_3 e^{-3t}\left(\vec{\theta} + t\vec{\eta} + \frac{t^2}{2}\vec{\xi}\right)$$

Case 3

Case 3: we have λ, associated with eigenvectors $\vec{\xi}_1, \vec{\xi}_2 \Rightarrow$ two solutions $e^{\lambda t}\vec{\xi}_1, e^{\lambda t}\vec{\xi}_2$.

To construct third solution, consider $\vec{x}^{(3)} = e^{\lambda t}(\vec{\eta} + t\vec{\beta})$ where $\vec{\beta}$ is an eigenvector, so it's a linear combination of $\vec{\xi}_1, \vec{\xi}_2$. $\vec{\eta}$ satisfies

$$\begin{cases} (A - \lambda I)\vec{\eta} = \vec{\beta}, \\ (A - \lambda I)^2\vec{\eta} = 0, \end{cases}$$

$\vec{\eta}$ is a generalized eigenvector.

Example

$$\vec{x}' = \begin{pmatrix} 4 & 6 & -15 \\ 1 & 3 & -5 \\ 1 & 2 & -4 \end{pmatrix} \vec{x}$$

Sol: $\det(A - \lambda I) = (1 - \lambda)^3 = 0 \Rightarrow \lambda = 1$. AM 3

A-I$= \begin{pmatrix} 3 & 6 & -15 \\ 1 & 2 & -5 \\ 1 & 2 & -5 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$$\vec{\xi}_1 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{\xi}_2 = \begin{pmatrix} 5 \\ 0 \\ 1 \end{pmatrix}$$

Choose $\vec{\beta} = \vec{\xi}_1 + \vec{\xi}_2$, and solve $(A - I)\vec{\eta} = \vec{\xi}_1 + \vec{\xi}_2$

$$\left(\begin{array}{ccc|c} 3 & 6 & -15 & 3 \\ 1 & 2 & -5 & 1 \\ 1 & 2 & -5 & 1 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 2 & -5 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \quad \vec{\eta} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Example

Set $\vec{\beta} = (A - I)\vec{\eta} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$ The general sol is

$$\vec{x}(t) = c_1 e^t \vec{\xi}_1 + c_2 e^t \vec{\xi}_2 + c_3 e^t (\vec{\eta} + t\vec{\beta})$$

Remark

If choose $\vec{\beta} = \vec{\xi}_1$, *and solve* $(A - I)\vec{\eta} = \vec{\xi}_1$

$$\left(\begin{array}{ccc|c} 3 & 6 & -15 & -2 \\ 1 & 2 & -5 & 0 \\ 1 & 2 & -5 & 1 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 2 & -5 & 1 \\ 1 & 2 & -5 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \quad \textit{no sol.}$$

Similarly. $(A - I)\vec{\eta} = \vec{\xi}_2$ *has no solution.*

You cannot choose $\vec{\beta} = \vec{\xi}_1$ *or* $\vec{\xi}_2$ *as previous case. It may not yield a solution of* $\vec{\eta}$.

Example

$$\vec{x}' = \begin{pmatrix} 4 & 6 & -15 \\ 1 & 3 & -5 \\ 1 & 2 & -4 \end{pmatrix} \vec{x}$$

$\lambda = 1$. AM=3 $\qquad \vec{\xi}_1 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{\xi}_2 = \begin{pmatrix} 5 \\ 0 \\ 1 \end{pmatrix}$

Method2: Start from $(A - I)^2 \vec{\eta} = 0$ check

$$(A - I)^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

so $\vec{\eta}$ is arbitrary and should be chosen as independent of $\vec{\xi}_1, \vec{\xi}_2$, choose $\vec{\eta} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. Then set $\vec{\beta} = (A - I)\vec{\eta} = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}$

So general sol is

$$\vec{x}(t) = c_1 e^t \vec{\xi}_1 + c_2 e^t \vec{\xi}_2 + c_3 e^t (\vec{\eta} + t\vec{\beta})$$

General & systematical approach for "repeated case"

$$\vec{x}' = A\vec{x},$$

Suppose λ is an eigenvalue with algebraic multiplicity m. Expect solution in the form:

$$\begin{aligned}
\vec{x} =& e^{\lambda t}\left(\vec{r}_0 + t\vec{r}_1 + \frac{t^2}{2!}\vec{r}_2 + \cdots + \frac{t^{m-1}}{(m-1)!}\vec{r}_{m-1}\right) \\
\Rightarrow \vec{x}' - A\vec{x} =& e^{\lambda t}\left(\vec{r}_1 - (A-\lambda I)\vec{r}_0\right) \\
&+ te^{\lambda t}\left(\vec{r}_2 - (A-\lambda I)\vec{r}_1\right) \\
&+ \frac{t^2}{2!}e^{\lambda t}\left(\vec{r}_3 - (A-\lambda I)\vec{r}_2\right) \\
&+ \cdots \\
&+ \frac{t^{m-2}}{(m-2)!}e^{\lambda t}\left(\vec{r}_{m-1} - (A-\lambda I)\vec{r}_{m-2}\right) \\
&+ \frac{t^{m-1}}{(m-1)!}e^{\lambda t}\left(-(A-\lambda I)\vec{r}_{m-1}\right)
\end{aligned}$$

General & systematical approach for "repeated case"

In order that $\vec{x}' - A\vec{x} = 0$, we require:

$$\begin{cases} \vec{r}_1 = (A-\lambda I)\vec{r}_0 \\ \vec{r}_2 = (A-\lambda I)\vec{r}_1 \\ \vdots \\ \vec{r}_{m-1} = (A-\lambda I)\vec{r}_{m-2} \\ 0 = (A-\lambda I)\vec{r}_{m-1} \end{cases} \Leftrightarrow \begin{cases} \vec{r}_1 = (A-\lambda I)\vec{r}_0 \\ \vec{r}_2 = (A-\lambda I)^2\vec{r}_0 \\ \vdots \\ \vec{r}_{m-1} = (A-\lambda I)^{m-1}\vec{r}_0 \\ 0 = (A-\lambda I)^m\vec{r}_0 \end{cases}$$

$$\Rightarrow \begin{cases} (A-\lambda I)\vec{r}_{m-1} = 0, & \vec{r}_{m-1} : \text{eigenvector} \\ (A-\lambda I)^2\vec{r}_{m-2} = 0, & \vec{r}_{m-2} : \text{generalized eigenvector of rank 2.} \\ \vdots & \vdots \\ (A-\lambda I)^{m-1}\vec{r}_1 = 0, & \vec{r}_1 = \text{ generalized eigenvector of rank } m-1. \\ (A-\lambda I)^m\vec{r}_0 = 0 & \vec{r}_0 = \text{ generalized eigenvector of rank } m. \end{cases}$$

$\{\vec{r}_0, \vec{r}_1, \cdots, \vec{r}_{m-1}\}$ is called a chain of generalized eigenvectors

General & systematical approach for "repeated case"

From the theory of linear algebra, if λ is an eigenvalue of A with algebraic multiplicity m, then$(A-\lambda I)^m \vec{r}_0 = 0$ has m linearly independent solutions:$\vec{r}_0^{(1)}, \vec{r}_0^{(2)}, \cdots, \vec{r}_0^{(m)}$

Then we construct the chains of generalized eigenvectors using above :

$$\begin{cases} \vec{r}_0^{(1)}, \\ \vec{r}_1^{(1)} = (A-\lambda I)\vec{r}_0^{(1)}, \\ \vec{r}_2^{(1)} = (A-\lambda I)r_1^{(1)}, \\ \vdots \\ \vec{r}_{m-1}^{(1)} = (A-\lambda I)\vec{r}_{m-2}^{(1)}. \end{cases} \quad \begin{cases} \vec{r}_0^{(2)}, \\ \vec{r}_1^{(2)} = (A-\lambda I)\vec{r}_0^{(2)}, \\ \vec{r}_2^{(2)} = (A-\lambda I)r_1^{(2)}, \\ \vdots \\ \vec{r}_{m-1}^{(2)} = (A-\lambda I)\vec{r}_{m-2}^{(2)}. \end{cases} \quad \begin{cases} \vec{r}_0^{(m)}, \\ \vec{r}_1^{(m)} = (A-\lambda I)\vec{r}_0^{(m} \\ \vec{r}_2^{(m)} = (A-\lambda I)r_1^{(m} \\ \vdots \\ \vec{r}_{m-1}^{(m)} = (A-\lambda I)\vec{r}_{m-} \end{cases}$$

Correspondingly, we have m solutions:

$$\vec{x}^{(1)} = e^{\lambda t}\left(\vec{r}_0^{(1)} + t\vec{r}_1^{(1)} + \frac{t^2}{2!}\vec{r}_2^{(1)} + \cdots + \frac{t^{m-1}}{(m-1)!}\vec{r}_{m-1}^{(1)}\right)$$

$$\vec{x}^{(2)} = e^{\lambda t}\left(\vec{r}_0^{(2)} + t\vec{r}_1^{(2)} + \frac{t^2}{2!}\vec{r}_2^{(2)} + \cdots + \frac{t^{m-1}}{(m-1)!}\vec{r}_{m-1}^{(2)}\right)\cdots$$

$$\vec{x}^{(m)} = e^{\lambda t}\left(\vec{r}_0^{(m)} + t\vec{r}_1^{(m)} + \frac{t^2}{2!}\vec{r}_2^{(m)} + \cdots + \frac{t^{m-1}}{(m-1)!}\vec{r}_{m-1}^{(m)}\right)$$

Example

$$\vec{x}' = \begin{pmatrix} -7 & -5 & -3 \\ 2 & -2 & -3 \\ 0 & 1 & 0 \end{pmatrix} \vec{x}$$

Sol: $\det(A - \lambda I) = -(\lambda + 3)^3 = 0 \quad \Rightarrow \lambda = -3$. AM is 3 . By direct calculation,

$$(A + 3I)^3 = \begin{pmatrix} -4 & -5 & -3 \\ 2 & 1 & -3 \\ 0 & 1 & 3 \end{pmatrix}^3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Therefore, the linearly independent solution to

$$(A + 3I)^3 \vec{r}_0 = 0$$

can be chosen as: $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$$\begin{cases} \vec{r}_0^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \\ \vec{r}_1^{(1)} = (A+3I)\vec{r}_0^{(1)} = \begin{pmatrix} -4 \\ 2 \\ 0 \end{pmatrix}, \\ \vec{r}_2^{(1)} = (A+3I)r_1^{(1)} = \begin{pmatrix} 6 \\ -6 \\ 2 \end{pmatrix}, \end{cases} \quad \begin{cases} \vec{r}_0^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \\ \vec{r}_1^{(2)} = (A+3I)\vec{r}_0^{(1)} = \begin{pmatrix} -5 \\ 1 \\ 1 \end{pmatrix}, \\ \vec{r}_2^{(2)} = (A+3I)r_1^{(1)} = \begin{pmatrix} 12 \\ -12 \\ 4 \end{pmatrix}, \end{cases}$$

$$\begin{cases} \vec{r}_0^{(3)} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \\ \vec{r}_1^{(3)} = (A+3I)\vec{r}_0^{(1)} = \begin{pmatrix} -3 \\ -3 \\ 3 \end{pmatrix}, \\ \vec{r}_2^{(3)} = (A+3I)r_1^{(1)} = \begin{pmatrix} 18 \\ -18 \\ 6 \end{pmatrix}, \end{cases}$$

Example

$$\vec{x}^{(1)} = e^{-3t}\left(\vec{r}_0^{(1)} + t\vec{r}_1^{(1)} + \frac{t^2}{2}\vec{r}_2^{(1)}\right) = e^{-3t}\begin{pmatrix} 1-4t+3t^2 \\ 2t-3t^2 \\ t^2 \end{pmatrix}$$

$$\vec{x}^{(2)} = e^{-3t}\left(\vec{r}_0^{(2)} + t\vec{r}_1^{(2)} + \frac{t^2}{2}\vec{r}_2^{(2)}\right) = e^{-3t}\begin{pmatrix} -5t+6t^2 \\ 1+t-6t^2 \\ t+2t^2 \end{pmatrix}$$

$$\vec{x}^{(3)} = e^{-3t}\left(\vec{r}_0^{(3)} + t\vec{r}_1^{(3)} + \frac{t^2}{2}\vec{r}_2^{(3)}\right) = e^{-3t}\begin{pmatrix} -3t+9t^2 \\ -3t-9t^2 \\ 1+3t+3t^2 \end{pmatrix}$$

General sol: $\vec{x} = c_1\vec{x}^{(1)} + c_2\vec{x}^{(2)} + c_3\vec{x}^{(3)}$.

Complex eigenvalue

Denote $\lambda = \alpha + i\beta, \quad \vec{\xi} = \vec{u} + i\vec{v}$ be the complex eigenvalue and eigenvector. We can check that $\lambda' = \alpha - i\beta, \quad \vec{\xi'} = \vec{u} - i\vec{v}$ is also a eigenvalue and corresponding eigenvector.
Complex valued solution is

$$e^{(\alpha+i\beta)t}(\vec{u} + i\vec{v}) = e^{\alpha t}(\cos\beta t \cdot \vec{u} - \sin\beta t \cdot \vec{v}) + ie^{\alpha t}(\sin\beta t \cdot \vec{u} + \cos\beta t \cdot \vec{v})$$

The real part and the imaginary part are both solutions. Hence the corresponding real value solution is:

$$\begin{cases} \vec{x}^{(1)} = e^{\alpha t}(\cos\beta t \cdot \vec{u} - \sin\beta t \cdot \vec{v}) \\ \vec{x}^{(2)} = e^{\alpha t}(\sin\beta t \cdot \vec{u} + \cos\beta t \cdot \vec{v}) \end{cases}$$

Example

$$\vec{x}' = \begin{pmatrix} -3 & -2 \\ 4 & 1 \end{pmatrix} \vec{x}$$

Sol: $|A - \lambda I| = \begin{vmatrix} -3-\lambda & -2 \\ 4 & 1-\lambda \end{vmatrix} = \lambda^2 + 2\lambda + 5 = 0,$

$\lambda_1 = -1 + 2i, \quad \lambda_2 = -1 - 2i.$

For $\lambda_1 = -1 + 2i = \alpha + \beta i$,

$$A - \lambda_1 I = \begin{pmatrix} -2-2i & -2 \\ 4 & 2-2i \end{pmatrix} \to \begin{pmatrix} 1+i & 1 \\ 0 & 0 \end{pmatrix}$$

Take $\vec{\xi}_1 = \binom{-1}{1+i} = \binom{-1}{1} + i\binom{0}{1} = \vec{u} + i\vec{v}$. so

$$\vec{x}^{(1)} = e^{-t}\cos 2t \begin{pmatrix} -1 \\ 1 \end{pmatrix} - e^{-t}\sin 2t \begin{pmatrix} 0 \\ 1 \end{pmatrix} = e^{-t} \begin{pmatrix} -\cos 2t \\ \cos 2t - \sin 2t \end{pmatrix}$$

$$\vec{x}^{(2)} = e^{-t}\sin 2t \begin{pmatrix} -1 \\ 1 \end{pmatrix} + e^{-t}\cos 2t \begin{pmatrix} 0 \\ 1 \end{pmatrix} = e^{-t} \begin{pmatrix} -\sin 2t \\ \sin 2t + \cos 2t \end{pmatrix}$$

and $\vec{x} = c_1\vec{x}^{(1)} + c_2\vec{x}^{(2)}$.

Summary

For constant coefficient homogeneous system: $\vec{x}' = A\vec{x}$.
(i)Enough eigenvector: $\lambda_1, \cdots, \lambda_n$ (may be equal), if there are n linearly independent eigenvectors: $\vec{\xi}_1, \cdots, \vec{\xi}_n$

$$\Rightarrow \left\{ e^{\lambda_1 t}\vec{\xi}_1, \cdots, e^{\lambda_n t}\vec{\xi}_n \right\}$$

(ii) If there are not n linearly independent eigenvectors: use generalized eigenvectors to find more solutions. eg:

$$\left.\begin{array}{l} \text{eigenvector } \vec{\xi} : (A-\lambda I)\vec{\xi} = 0. \\ \text{generalized eigenvector } \vec{\eta} : (A-\lambda I)\vec{\eta} = \vec{\xi} \end{array}\right\} \Rightarrow e^{\lambda t}\vec{\xi}, e^{\lambda t}(t\vec{\xi} + \vec{\eta}).$$

If complex eigenvalues occur:find the complex valued solution following (i) or (ii), then take the real part and imaginary part. eg: for case (i):
$\lambda = \alpha + i\beta, \vec{\xi} = \vec{u} + i\vec{v}$,

$$\Rightarrow e^{\alpha t}(\cos\beta t\vec{u} - \sin\beta t\vec{v}), e^{\alpha t}(\sin\beta t\vec{u} + \cos\beta t\vec{v})$$